Technical specification template: Power Cables

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for power cables for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 60502 series (power cables with extruded insulation)
- IEC 60287 series (cable current rating calculation)
- IEC 60364 series (installations), including current-carrying capacity
- Local fire performance requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Installation method (basis of the rating)	Single-core cables in trefoil on ventilated cable tray in air	PRELIMINARY_ASSUMPTION
Conductor operating temperature	90.0 degC	UNVERIFIED_PLACEHOLDER. Of the chosen insulation
Voltage drop limit	3.0 %	PRELIMINARY_ASSUMPTION. Project criterion
Cable sizes and runs	See the cable sizing workbook: illustrative selections only	UNVERIFIED_PLACEHOLDER
Tabulated current rating per run	[From the cable standard or manufacturer for the installation method; never assumed]	Pending
Fault withstand	[Conductor area to survive the calculated fault for the clearing time]	NOT CALCULATED

5. Technical requirements

5.1 Each cable shall carry its design current after applying the grouping, ambient and installation derating factors, using a tabulated rating for the actual installation method.

5.2 The cable shall keep the voltage drop within the project limit at the design current.

5.3 The conductor shall withstand the calculated fault current for the protection clearing time (adiabatic or manufacturer data).

5.4 Insulation, sheath, armour, fire performance (flame propagation, smoke and halogen emission) and identification: [project to define].

5.5 Parallel runs shall be of equal length and layout with the current sharing verified; bending radius and pulling tensions per the manufacturer.

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
cable_data.lv_reactance_ohm_per_km	LV cable reactance	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	UNVERIFIED_PLACEHOLDER
cable_data.mv_reactance_ohm_per_km	MV cable reactance	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient	UNVERIFIED_PLACEHOLDER

		factors	
cable_data.conductor_operating_temperature	Conductor temperature for resistance	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	UNVERIFIED_PLACEHOLDER
distribution.CAB-MV-A.tabulated_rating_a	CAB-MV-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING
distribution.CAB-GEN-A.tabulated_rating_a	CAB-GEN-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING
distribution.CAB-UPS-IN-A.tabulated_rating_a	CAB-UPS-IN-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING
distribution.CAB-UPS-OUT-A.tabulated_rating_a	CAB-UPS-OUT-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING
distribution.CAB-MECH-A.tabulated_rating_a	CAB-MECH-A: tabulated_rating_a	Cable data sheet or the standard's current-rating table for the installation method, with grouping and ambient factors	MISSING
short_circuit.assumed_clearing_time_s	Assumed fault clearing time for cable and busduct withstand	Protection coordination study result and protective device time-current curves	UNVERIFIED_PLACEHOLDER

7. Inspection and quality requirements

7.1 Review of type-test certificates and drum test records.

7.2 Inspection of drum condition, markings and length.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Routine tests per the standard: conductor resistance, voltage test, partial discharge for MV cables.

8.2 Witnessing: [project to define].

8.3 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Insulation resistance and continuity; phasing.

9.2 MV cable test per the specification [project to define: very low frequency or DC withstand and partial discharge].

9.3 Verification of installed layout against the rating basis (spacing, grouping, burial or tray).

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.